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Logical Connectives



Learning Objectives

At the end of the lesson, the students must be able to:

- Define key logical connectives such as "and" ($\wedge$), "or" ($\vee$), "not" ($\neg$), "if... then" ($\rightarrow$), and "if and only if" ($\leftrightarrow$).
- Translate verbal statements into formal logical expressions using connectives.
- Construct truth tables for compound logical statements involving one or more logical connectives.



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What is logical connectives?

- It is a symbol which is used to connect two or more proposition or predicate logics in such a manner that resultant logic depends only on the input logics and the meaning of the connectives used.



Five Connectives

- OR ($\vee$)
- AND ($\wedge$)
- Negation/NOT ($\neg$)
- Implication/If-then ($\rightarrow$)
- If and only if ($\leftrightarrow$)



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OR ($\vee$)

The OR operation of two propositions A and B (written as $A \vee B$) is true if at least any of the propositional variable A or B is true.



Truth Table

A	B	$A \vee B$
True	True	True
True	False	True
False	True	True
False	False	False



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AND ($\wedge$)

The AND operation of two propositions A and B (written as $A \wedge B$) is true if both the propositional variable A or B is true.



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Truth Table

A	B	$A \vee B$
True	True	True
True	False	False
False	True	False
False	False	False



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Negation (-)

The negation of a propositions A (written as $\neg A$) is false when A is true and is true when A is false.



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Truth Table

A	-A
True	False
False	True



Implication/If- then ($\rightarrow$)

An implication $A \rightarrow B$ is the propositions A "If A, then B". It is false if A is true and B is false. The rest cases are true.



Truth Table

A	B	$A \rightarrow B$
True	True	True
True	False	False
False	True	True
False	False	True



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If and only if
($\longleftrightarrow$)

$A \longleftrightarrow B$ is bi-conditional logical
connective which is true when p and q
are same, i.e both are false or both are
true.



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Truth Table

A	B	$A \rightarrow B$
True	True	True
True	False	False
False	True	False
False	False	True



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REFERENCE:

<https://www.tutorialspoint.com/mathematical-logical-connectives>